

**Placeholder University**  
**Placeholder Doctoral School**  
*Information Sciences & Technologies*

# Doctoral Thesis

Presented by:  
**Fourth STUDENT**

## A PLACEHOLDER TITLE FOR A PLACEHOLDER THESIS

Defended publicly on **September 30<sup>th</sup>, 2026** before the jury composed of:

**Firstname REVIEWERONE**, Senior Researcher, HDR (Reviewer)  
Placeholder Laboratory, Placeholder University

**Firstname REVIEWERTWO**, Professor, HDR (Reviewer)  
Placeholder Institute

**Firstname EXAMINERONE**, Researcher, HDR (Examiner)  
Placeholder Institute

**Firstname LASTNAME**, Associate Professor, HDR (Supervisor)  
Placeholder Laboratory

**Second COAUTHOR**, Researcher (Co-supervisor)  
Placeholder Laboratory



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## ACKNOWLEDGEMENTS



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This chapter carries no number, and does not appear in the mini table of contents of the chapters that follow. It can still be referred to like any other one: `\nameref{phd_acknowledgements}` gives [Acknowledgements](#).



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# CHAPTER I

## INTRODUCTION



A PhD manuscript uses exactly the same commands as an HDR one: only the `phd` option of the document class differs.

This chapter introduces the placeholder problem addressed in this thesis, and outlines the rest of the document.

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## I.1 Context

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The rest of this document uses ML vocabulary, and in particular the Convolutional Neural Network (CNN). Note that CNNs is a separate entry of `acronyms.tex`, whose only purpose is to provide a plural tooltip: its full name is Convolutional Neural Networks, and it does not appear in the list of acronyms. Recall that State Of The Art (SOTA) results in this placeholder field are due to [AMET *et al.*, 2017 .

## I.2 Problem Statement

### Problem 1 Placeholder problem

Given a set of placeholder observations  $\{\mathbf{x}_i\}_{i=1}^N$  and their placeholder targets  $\{\mathbf{y}_i\}_{i=1}^N$ , find a placeholder model  $f_\theta$  that generalizes to unseen placeholder observations.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Problem 1 is addressed in Chapter III, after the review of Chapter II.

## I.3 Outline

Chapter II reviews the placeholder literature. Chapter III presents our contribution, evaluated in Section III.2. Chapter IV concludes, and Appendix A gathers the proofs. The document opens on the Acknowledgements and closes on the Bibliography and the list of Acronyms: these three chapters are unnumbered, and are therefore recalled by their title rather than by a number.



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## CHAPTER II

# STATE OF THE ART



This chapter reviews the placeholder literature, and positions the contribution of [Chapter III](#) with respect to it.

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## II.1 Early Approaches

Lorem ipsum dolor sit amet, consectetur adipiscing elit [IPSUM *et al.*, 1998 ]. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua, as summarized in the textbook of [TEMPOR *et al.*, 2009 .

### Definition 1 Placeholder representation

A *placeholder representation* of an observation  $x$  is any vector  $z = g(x)$  such that  $z$  carries no more information than  $x$ .

## II.2 Modern Approaches

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat [AMET *et al.*, 2017 ]. CNNs are now trained on a single GPU, which was not the case for the earlier approaches of Section II.1. Figure 1 summarizes the pipeline they all share.



Figure 1 An overview of the placeholder pipeline shared by modern approaches. The same figure is reused in Chapter III, which is perfectly fine as long as the labels differ.

## II.3 Summary

Table 1 compares the approaches discussed above. None of them addresses Problem 1 directly, which motivates the contribution of Chapter III.

| Approach                                                                                                           | Placeholder | Scalable | Interpretable |
|--------------------------------------------------------------------------------------------------------------------|-------------|----------|---------------|
| [IPSUM <i>et al.</i> , 1998  ]  | ✓           | ✗        | ✓             |
| [TEMPOR <i>et al.</i> , 2009  ] | ✓           | ✗        | ✗             |
| [AMET <i>et al.</i> , 2017  ]   | ✓           | ✓        | ✗             |
| <b>Ours</b>                                                                                                        | ✓           | ✓        | ✓             |

Table 1 A comparison of placeholder approaches. Checkmarks are placeholders too.

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## CHAPTER III

### A PLACEHOLDER CONTRIBUTION



This chapter presents the contribution of this thesis, and is the one that demonstrates most of the environments provided by the template.

It answers [Problem 1](#), stated in [Chapter I](#).

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### III.1 Method

We look for a placeholder model  $f_\theta$  minimizing the objective of Equation 1, where  $\lambda > 0$  controls the strength of the placeholder regularizer.

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N} \sum_{i=1}^N \|f_\theta(\mathbf{x}_i) - \mathbf{y}_i\|_2^2}_{\text{placeholder fidelity}} + \lambda \underbrace{\|\theta\|_2^2}_{\text{placeholder regularizer}} \quad (1)$$

**Proposition 1** Existence of a placeholder minimizer

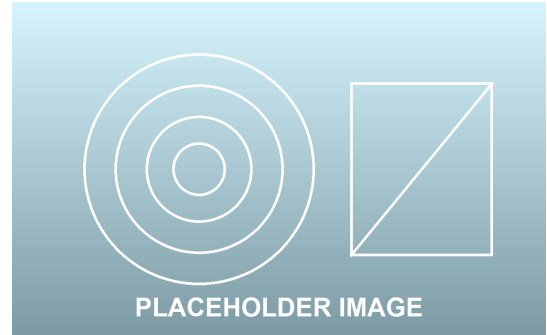
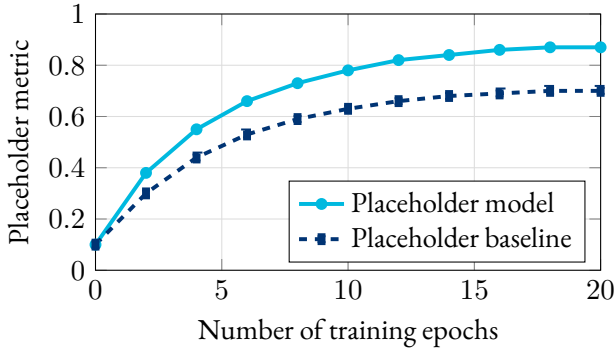
For every  $\lambda > 0$ , the objective of Equation 1 admits at least one minimizer  $\theta^*$ .

The proof of Proposition 1 is deferred to Appendix A, so as not to interrupt the exposition. An unnumbered equation is obtained with the starred variant, which is convenient for intermediate steps:

$$\nabla_\theta \mathcal{L}(\theta) = \frac{2}{N} \sum_{i=1}^N (f_\theta(\mathbf{x}_i) - \mathbf{y}_i) \nabla_\theta f_\theta(\mathbf{x}_i) + 2\lambda\theta$$

### III.2 Results

We evaluate the method of Section III.1 on placeholder datasets. Figure 2 reports the learning curves, and Table 2 the final scores.



(b) A qualitative placeholder example.

(a) Placeholder metric along training, for our model and the baseline of [AMET et al., 2017].

Figure 2 Placeholder results. (a) reports the metric along training, while (b) shows a qualitative example.

Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. These results were partly published in [LASTNAME et al., 2021] and [COAUTHOR et al., 2024].

| Dataset       | [AMET <i>et al.</i> , 2017  | Ours               |
|---------------|--------------------------------------------------------------------------------------------------------------|--------------------|
| Placeholder-A | 0.70 ± 0.02                                                                                                  | <b>0.87 ± 0.01</b> |
| Placeholder-B | 0.61 ± 0.03                                                                                                  | <b>0.79 ± 0.02</b> |
| Placeholder-C | 0.55 ± 0.04                                                                                                  | <b>0.68 ± 0.03</b> |

Table 2 Final placeholder scores, averaged over five placeholder runs. Best results in bold.

### III.3 Discussion

 Limitations

An unnumbered box, obtained with `boxenv*`, is handy for remarks that do not need to be cross-referenced. The placeholder method has placeholder limitations, which are discussed in [Chapter IV](#).



## CHAPTER IV

# CONCLUSION & PERSPECTIVES



A short closing chapter, summarizing the contribution of [Chapter III](#) and listing the directions it opens.

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## IV.1 Summary

This thesis addressed [Problem 1](#), introduced in [Chapter I](#). After the review of [Chapter II](#), [Chapter III](#) proposed a placeholder method, whose objective is given in [Equation 1](#) and whose results are reported in [Table 2](#).

## IV.2 Perspectives

- Extending the placeholder method to placeholder data of higher dimension.
- Reducing the GPU cost of placeholder training, so that the approach scales beyond placeholder datasets.
- Studying the interpretability of the placeholder representations of [Definition 1](#).

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# APPENDIX A

## PROOFS



Deferring proofs to an appendix keeps the main text readable.

The `proof` environment takes the label of the statement it proves, and displays a clickable reference to it in its title.

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## A.1 Existence of a Placeholder Minimizer

### Proof of Proposition 1

The objective of Equation 1 is continuous in  $\theta$ , and coercive as soon as  $\lambda > 0$ , since  $\|\theta\|_2^2 \rightarrow \infty$  when  $\|\theta\|_2 \rightarrow \infty$ . A continuous coercive function on  $\mathbb{R}^d$  attains its infimum, hence the existence of a minimizer  $\theta^*$ .  $\square$

## A.2 A Proof Without a Statement

The mandatory argument of proof can also be left empty, in which case the box is simply titled “Proof”.

### Proof

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# BIBLIOGRAPHY



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Firstname LASTNAME, Second COAUTHOR, Third CONTRIBUTOR (2021). **“Sed Do Eiusmod Tempor: Learning Placeholder Representations with CNNs”**. *Journal of Placeholder Research* (vol. 12, nb. 3, pp. 42–58).

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 —  6, 8, 10, 11

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## Online Resources Cited in this Document

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 August 19<sup>th</sup>, 2026

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 August 19<sup>th</sup>, 2026

**“A Placeholder Online Resource”**. *Placeholder Organization*.



## ACRONYMS



|      |       |                                                               |
|------|-------|---------------------------------------------------------------|
| AI   | ..... | <b>A</b> rtificial <b>I</b> ntelligence                       |
| BSc  | ..... | <b>B</b> achelor of <b>S</b> cience                           |
| CNN  | ..... | <b>C</b> onvolutional <b>N</b> eural <b>N</b> etwork          |
| GNU  | ..... | <b>G</b> NU's <b>N</b> ot <b>U</b> nix                        |
| GPL  | ..... | <b>G</b> eneral <b>P</b> ublic <b>L</b> icense                |
| GPU  | ..... | <b>G</b> raphics <b>P</b> rocessing <b>U</b> nit              |
| HDR  | ..... | <b>H</b> abilitation à <b>D</b> iriger des <b>R</b> echerches |
| ML   | ..... | <b>M</b> achine <b>L</b> earning                              |
| MSc  | ..... | <b>M</b> aster of <b>S</b> cience                             |
| PDF  | ..... | <b>P</b> ortable <b>D</b> ocument <b>F</b> ormat              |
| PhD  | ..... | <b>P</b> hilosophiæ <b>D</b> octor                            |
| SOTA | ..... | <b>S</b> tate <b>O</b> f <b>T</b> he <b>A</b> rt              |





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